

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

```
# Load Iris Dataset
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
# Split the data
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
# Scale the data
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
# Create Neural Network
```

```
model = MLPClassifier(
    hidden_layer_sizes=(2, 2, 2),
    activation='relu',
    learning_rate_init=0.001,
```

```
max_iter=2000,  
random_state=42  
)  
# Train the model  
model.fit(X_train, y_train)  
# Prediction  
y_pred = model.predict(X_test)  
# Accuracy  
accuracy = accuracy_score(y_test, y_pred)  
print("Neural Network Analysis for Multi-Class Data")  
print("Learning Rate: 0.001")  
print("Activation Function: ReLU")  
print("Hidden Layers: 3")  
print("Hidden Neurons: 2")
```

```
.. Neural Network Analysis for Multi-Class Data  
   Learning Rate: 0.001  
   Activation Function: ReLU  
   Hidden Layers: 3  
   Hidden Neurons: 2  
   Accuracy: 0.6333333333333333
```